
AI-Powered Urban Air Quality Intelligence Platform

From Reactive Monitoring to Proactive, Evidence-Based Intervention

VAYU

Team: BugByte

Problem Statement: AI-Powered Urban Air Quality Intelligence for Smart City Intervention

Theme: Smart Cities / Environmental Intelligence / Geospatial Analytics / Public Health



1. Problem Understanding

India's air quality crisis extends well beyond Delhi. In 2024-25, Delhi averaged an AQI of 218, classified 'Poor' or worse for over 200 days, while Mumbai recorded dangerous AQI levels on more than 60 days in 2024. Kolkata averaged an AQI above 150 through large parts of winter, and even Bengaluru and Chennai historically considered cleaner cities have shown measurable deterioration as vehicle density and construction activity increase.

CPCB's National Air Quality data for 2024 shows that 24 of India's 50 most polluted cities are Tier 1 or Tier 2 urban centres, and The Lancet Planetary Health has estimated 1.67 million premature deaths annually in India from air pollution a burden that falls disproportionately on urban populations.

India has deployed over 900 Continuous Ambient Air Quality Monitoring Stations (CAAQMS) under the National Clean Air Programme, yet a 2024 CAG audit found that only 31% of cities with monitoring data had any actionable multi-agency response protocol linked to those readings.

The core gap: the data exists the intelligence layer to act on it does not. City administrations need geospatial attribution of pollution sources, hyperlocal predictive forecasting, and enforcement intelligence, combined in one system. That combination does not exist today.

2. Proposed Solution

VAYU is a unified AI-powered command center that fuses monitoring station data, satellite imagery, mobility feeds, meteorological forecasts, and geospatial land-use layers to move city administrators from reactive monitoring to proactive, evidence-based intervention. Rather than building a single-purpose tool, VAYU integrates all five intervention areas outlined in the problem statement into one connected platform:

- Geospatial Pollution Source Attribution Engine : attributes pollution by source category (traffic, construction, industry, waste burning) at ward/zone level with statistical confidence scores.
- Hyperlocal Predictive AQI Forecasting Agent : 24-72 hour AQI forecasts at 1km grid resolution using meteorological, traffic, and seasonal emission data.
- Enforcement Intelligence & Prioritisation Agent : correlates hotspots with registered emission sources and generates ranked, evidence-backed enforcement recommendations.
- Multi-City Comparative Intelligence Dashboard : benchmarks intervention effectiveness and compliance across cities so administrators can learn from what has worked elsewhere.
- Citizen Health Risk Advisory System : ward-level health risk alerts mapped against vulnerable populations, delivered via app, public display, and IVR in regional languages.

The unifying idea is actionability: every insight in VAYU is paired with a concrete next step a dispatch order, an advisory, a comparison rather than a static number on a dashboard.

3. System Architecture

VAYU follows a four-layer architecture: data ingestion, an AI/ML intelligence layer, shared geospatial services, and a presentation layer. This separation allows each intelligence agent (attribution, forecasting, enforcement, advisory) to be developed, trained, and scaled independently while sharing a common data backbone.

VAYU — System Architecture

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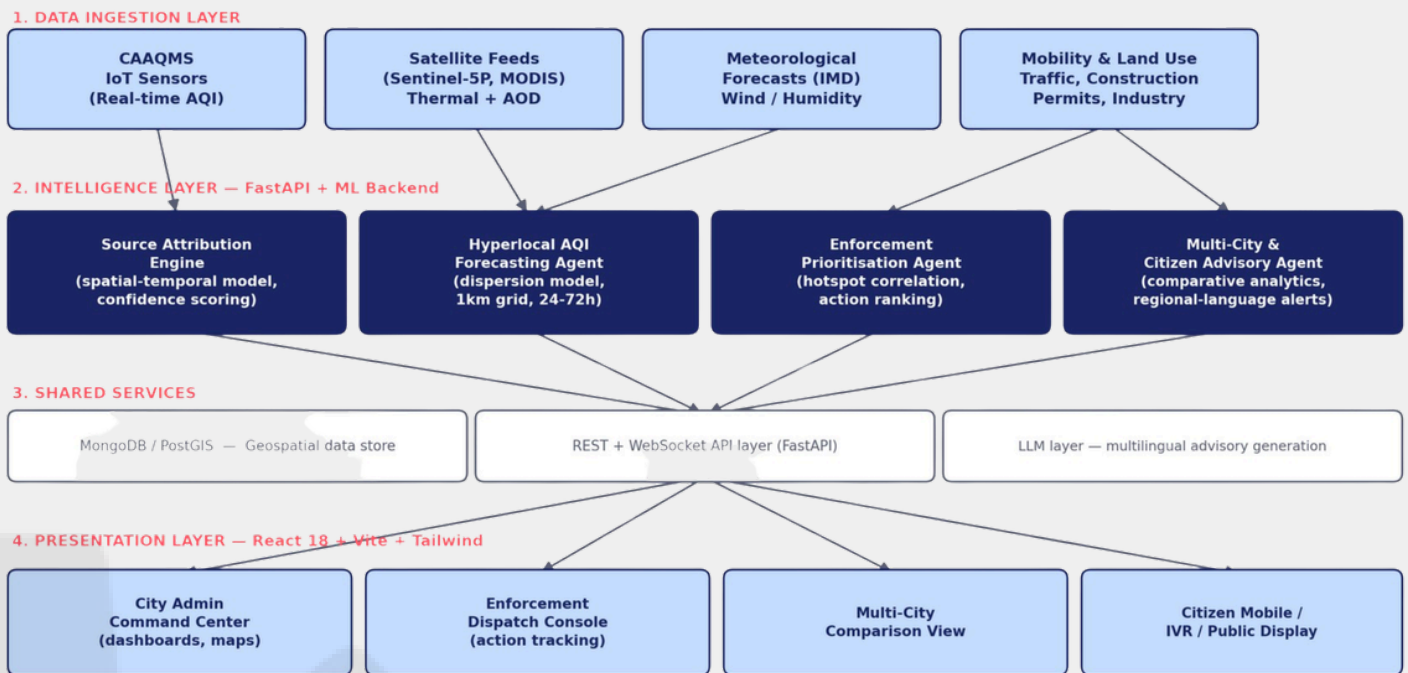


Figure 1: VAYU high-level system architecture

Data flows bottom-up conceptually but top-down in the diagram: raw sensor, satellite, weather, and mobility data is ingested and normalised, processed by the four intelligence agents, exposed through a common FastAPI/WebSocket layer backed by MongoDB/PostGIS for geospatial storage, and finally rendered across the React-based command center, dispatch console, comparison view, and citizen-facing channels.

4. Module Deep-Dive

4.1 Geospatial Pollution Source Attribution Engine

This module ingests spatial-temporal AQI readings and cross-references them against land-use maps, traffic density, active construction permits, industrial stack locations, and satellite-detected thermal anomalies (from Sentinel/MODIS). A multi modal model attributes the likely pollution source category at ward or zone level and outputs a statistical confidence score, giving administrators a defensible basis for action rather than a guess.

4.2 Hyperlocal Predictive AQI Forecasting Agent

The forecasting agent combines IMD meteorological forecasts, predicted traffic patterns, seasonal emission calendars (e.g. stubble burning, festival fireworks), and atmospheric dispersion modelling to project AQI 24-72 hours ahead at 1km grid resolution. This shifts city response from reactive advisories issued after AQI spikes to scheduled interventions for example, pre-emptively restricting construction activity ahead of a forecast pollution spike.

4.3 Enforcement Intelligence & Prioritisation Agent

This agent correlates live pollution hotspots with a registry of known emission sources industries, construction sites, waste-burning locations, and diesel fleet movement to produce a ranked list of enforcement actions. Each recommendation carries supporting geospatial evidence (satellite thermal signature, sensor trend, source registry match), so field inspectors are directed to the highest-impact locations first instead of patrolling reactively.

4.4 Multi-City Comparative Intelligence Dashboard

City administrators rarely have visibility into what has worked in comparable cities. This module tracks AQI trends, intervention effectiveness, and compliance metrics across multiple urban centres side by side, so a city piloting an intervention can see whether a similar measure succeeded elsewhere before committing resources.

4.5 Citizen Health Risk Advisory System

This module overlays forecast AQI against population vulnerability data hospitals, schools, outdoor worker zones, elderly populations to generate ward-level health risk alerts. Advisories are pushed through mobile apps, public displays, and IVR in regional languages (e.g. Kannada in Bengaluru, Tamil in Chennai), extending reach to citizens without smartphone access.

5. Technology Stack & Data Sources

Layer	Technology	Purpose
Frontend	React 18, Vite, Tailwind CSS	Command center, dashboards, maps, mobile-responsive UI
Backend / API	FastAPI (Python), WebSockets	Serving predictions, real-time updates, agent orchestration
ML / Modelling	Scikit-learn, TensorFlow/PyTorch	Source attribution classification, AQI forecasting models
Geospatial data store	MongoDB, PostGIS	Storing sensor, satellite-derived and zonal data
External data sources	CPCB CAAQMS API, Sentinel-5P, MODIS, IMD forecasts	Ground-truth AQI, satellite AOD/thermal data, weather forecasts
LLM layer	Anthropic Claude API	Multilingual citizen advisory generation, enforcement report drafting

6. Impact & Scalability

- Response time: reduces the signal to intervention gap by moving from manual, after the fact review to automated, ranked enforcement recommendations generated as hotspots emerge.
- Forecast accuracy: 24-72 hour hyperlocal forecasts (1km grid) evaluated against a persistence baseline using RMSE, enabling scheduled rather than reactive intervention.
- Multi-city scalability: the ingestion and modelling layers are city-agnostic onboarding a new city requires new CAAQMS station IDs and a land-use/zone boundary file, not a rebuild.
- Language coverage: citizen advisories are generated through the LLM layer per city in the locally dominant language, extending reach beyond English/Hindi dashboards.

7. Alignment with Judging Criteria

Criteria	Weight	How VAYU Addresses It
Innovation	25%	Single unified platform covering all five intervention areas from the problem statement, rather than one isolated feature.
Business Impact	25%	Every insight is paired with an action — dispatch, advisory, or comparison — directly answering the stated gap between data and intervention.
Technical Excellence	20%	Modular multi-agent architecture (FastAPI + ML models) with a clear path from prototype to production data sources.
Scalability	15%	City-agnostic ingestion layer; multi-city dashboard is designed for cross-city comparison from day one.
User Experience	15%	Purpose-built command-center UI for administrators, plus a separate simplified citizen-facing advisory experience in regional languages.

8. Team & Project Links

Team: BugByte

Team Leader:

- [Aarav Kumar](#)

Team Member:

- [Suvi Kumari](#)
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Project Resources

Live Application

Frontend (Web App):-

<https://vayu-weld.vercel.app/app>

Backend API:-

<https://vayu-server.onrender.com/>

AI Prediction Service:-

<https://vayu-ai-service.onrender.com/>

Source Code

GitHub Repository:-

<https://github.com/aarav12e/VAYU>

9. Research & Data References

The solution has been designed using publicly available environmental datasets, geospatial resources, and AI research.

Primary References

- Central Pollution Control Board (CPCB) – National Air Quality Monitoring Programme
- CPCB Continuous Ambient Air Quality Monitoring Stations (CAAQMS)
- India Meteorological Department (IMD)
- Sentinel-5P Satellite (European Space Agency)
- NASA MODIS Satellite Data
- OpenStreetMap
- Anthropic Claude API Documentation
- FastAPI Documentation
- MongoDB Documentation
- React Documentation